Transient / Steady State Response

We may check the responses for 3 characteristics

1- Quality of response

2- Accuracy of response (% error)

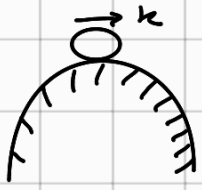
3- Stability

Differential equations: 1st $T\dot{x} + x = r$; 2nd $\ddot{x} + a_1\dot{x} + a_2 x = r$

$x(t) = x_h(t) + x_p(t)$

- $x_h(t)$: homogenous solution — due to I.C — Transient response
- $x_p(t)$: particular solution — due to external excitation

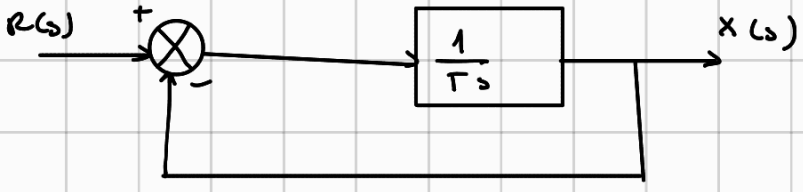
To check stability we have to check the homogen solution



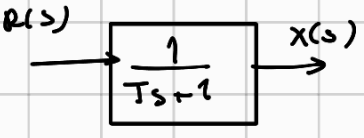
or it can be unstable due to resonance $r = \varepsilon \sin \omega t$

To get $x_n(t) \rightarrow \begin{cases} T\dot{x} + x = 0 \\ \ddot{x} + a_1\dot{x} + a_2 x = 0 \end{cases}$

First order system



$\rightarrow$



$T\frac{dx}{dt} + x = r$ initial condition $x(0) = ✓$

$RC\frac{dV_c}{dt} + V_c = V$

$T\dot{V}_c + V_c = V \rightarrow (Ts+1)V_c = V$ ← characteristic equation

$\mathcal{L}\{T[sX(s) - x_0]\} = R \rightarrow X(s)(Ts+1) = R(s) + Tx(0)$

$$\rightarrow X(s) = \underbrace{\frac{R(s)}{Ts+1}}_{\text{particular solut}} + \underbrace{\frac{Tx(0)}{Ts+1}}_{\text{homo sol.}}$$

if $R(s) = 0$ → transient solution

$T = RC$

$$X(s) = \frac{Tx(0)}{Ts+1} = Tx(0)\left(\frac{1/T}{s+1/T}\right) = \frac{x(0)}{s+\frac{1}{T}}$$

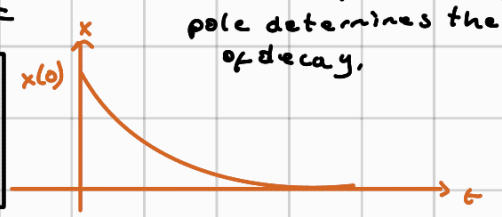
s_1 = position of pole determines the rate of decay.

$\downarrow L^{-1}$

$$x(t) = x_0 e^{-t/T} = x_0 e^{s_1 t}$$

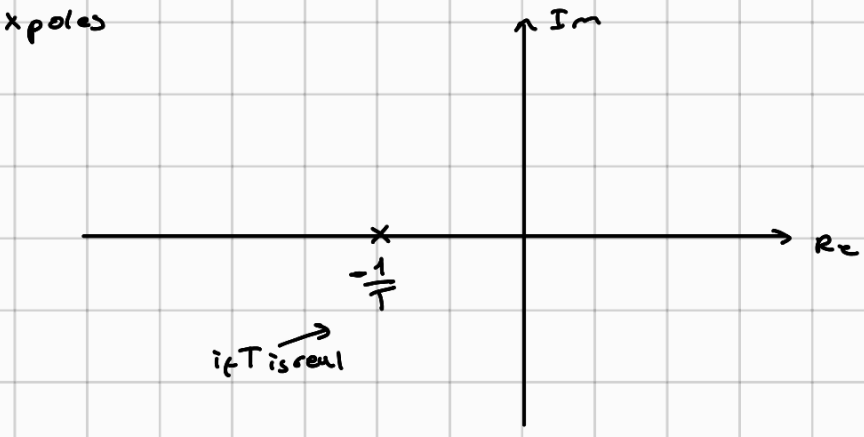
$$x(t) = \frac{x(0)}{e^{t/T}} = \frac{x(0)}{e^{|s_1|t}}$$

$|s_1| \uparrow$ $x(t)$ decrease faster



T: time constant of the system

characteristic equation : $Ts+1 \rightarrow s_1 = -\frac{1}{T}$ poles of the syten



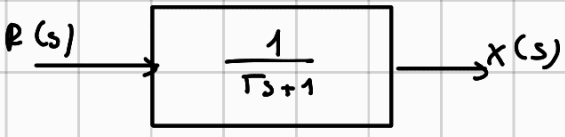
$T \uparrow \Rightarrow |s_1| \downarrow \Rightarrow$ closer to imaginary axis

$T \downarrow \Rightarrow |s_1| \uparrow \Rightarrow$ further from " "

First order systems

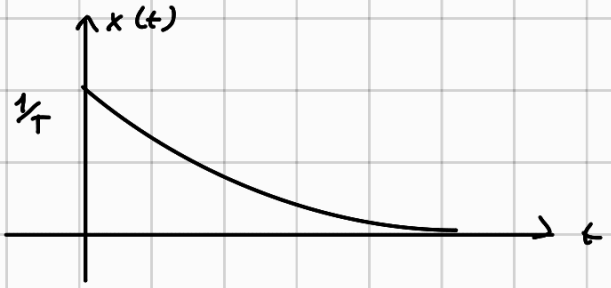
Assume that there i no intition condition

and $R(s) \neq 0$



case1) impulse input : $R(s) = 1$

$$X(s) = \frac{1}{Ts+1} = \frac{1/T}{s+\frac{1}{T}} \Rightarrow x(t) = \frac{1}{T} e^{-t/T} = |s_1| e^{-|s_1|t}$$



Case II) Step excitation

$r(t) = 1 \qquad t \geq 0$

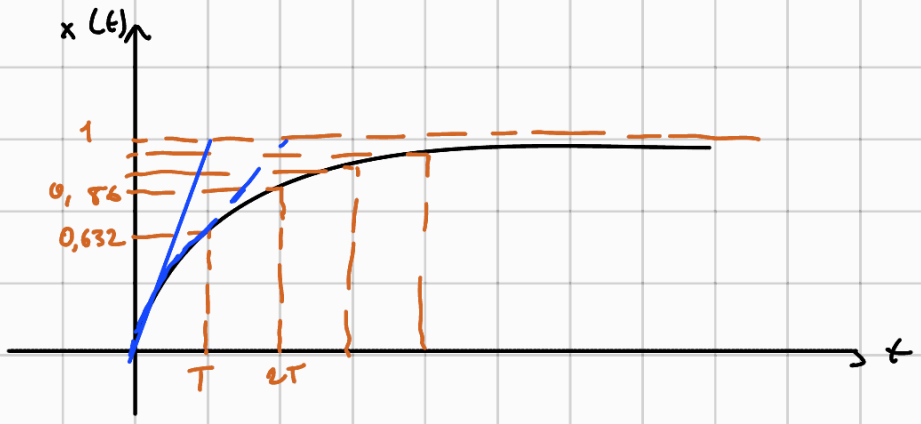
$R(s) = \frac{1}{s}$

$$X(s) = \frac{1}{s(Ts+1)} = \frac{A}{s} + \frac{B}{Ts+1} = \frac{1}{s} - \frac{\frac{1}{T}}{Ts+1} = \frac{1}{s} - \frac{1}{s+1/T}$$

$\xrightarrow{\mathcal{L}^{-1}} \quad x(t) = 1 - e^{-t/T}$

$$A = \frac{s}{s(Ts+1)}\Big|_{s=0} = 1$$

$$B = \frac{Ts+1}{s(Ts+1)}\Big|_{s=-\frac{1}{T}} = -T$$



at $t=0 \rightarrow x(0)=0$

t	$x(t)$
T	0,632
$2T$	0,86
$3T$	0,99
$4T$	0,98

63,2% of the final value

it have to be reasonable setting time

$$e(t) = r(t) - x(t) = 1 - 1 + e^{-t/T} = e^{-t/T}$$

derivative of response gives us the rate of change (slope of response)

$$\text{slope} = \frac{dx(t)}{dt} = \frac{1}{T} e^{-t/T}$$

at $t=0$, we get the max slope: $\left(\frac{1}{T}\right)$

$T\dot{x} + x = r$

only r has to be continuos

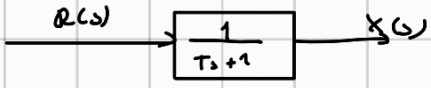
$\dddot{x} + a_1\ddot{x} + a_2\dot{x} = r$

x and $\dot{x}$ has to be continues

Impulse respone = $\frac{d}{dt}$ (step response)

1st order system

III) Ramp Respone



$r(t) = t$

$R(s) = \frac{1}{s^2}$

$X(s) = R(s)\frac{1}{Ts+1} = \frac{1}{s^2(Ts+1)}$

$s_{1,2} = 0$

$s_3 = -\frac{1}{T}$

$= \frac{A}{s^2} + \frac{B}{s} + \frac{C}{Ts+1}$

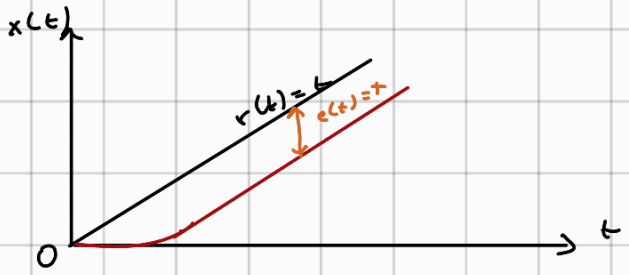
$A = s^2\left(\frac{1}{s^2(Ts+1)}\right)\Big|_{s=0} = 1$

$B = \frac{d}{dt}\left(s\frac{1}{s^2(Ts+1)}\right)\Big|_{s=0} = -T$

$C = (Ts+1)\frac{1}{s^2(Ts+1)}\Big|_{s=-\frac{1}{T}} = T^2$

$$X(s) = \frac{1}{s^2} - \frac{T}{s} + \frac{T^2}{Ts+1} = \frac{T}{s+\frac{1}{T}}$$

$$\mathcal{L}^{-1}\Big\} \; x(t) = t - T + Te^{-t/T}$$



$$\hat{x} = 1 - e^{-t/T}$$

$$e(t) = r(t) - x(t)$$

$$= t - (t - T + Te^{-t/T})$$

$$= T - Te^{-t/T}$$

⊙ $t = 0 \longrightarrow e(0) = 0$

$t \to \infty \Rightarrow e(\infty) = e_{ss} = T$

Accuracy is declined by 'T'.

Ramp response = $\int$ unit response = $\iint$ impulse response

$$\frac{d}{dt}(\text{ramp resp}) = \text{step resp}$$

$$\frac{d^2}{dt^2}(\text{ramp resp}) = \text{impulse resp}$$

2nd order response

RLC in series

$$\frac{V_c(s)}{V(s)} = \frac{1}{LCs^2 + RCs + 1} = \frac{1/LC}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$$

RLC in parallel

$$\frac{\Phi(s)}{I(s)} = \frac{1}{Cs^2 + \frac{s}{R} + \frac{1}{L}} = \frac{1/C}{s^2 + \frac{s}{RC} + \frac{1}{LC}}$$

Spring mass Damper System

$$\frac{X(s)}{R(s)} = \frac{1}{ms^2 + cs + k} = \frac{1/m}{s^2 + \frac{c}{m}s + \frac{k}{m}}$$

$$\Rightarrow \frac{\text{output}}{\text{input}} = \frac{X(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

ω_n = naturally frequency

ζ = zeta = damping ratio

no zeros but 2 poles are present

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$s_{1,2} = \frac{-2\zeta\omega_n \mp \sqrt{(2\zeta\omega_n)^2 - 4\omega_n^2}}{2}$$

$$= -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

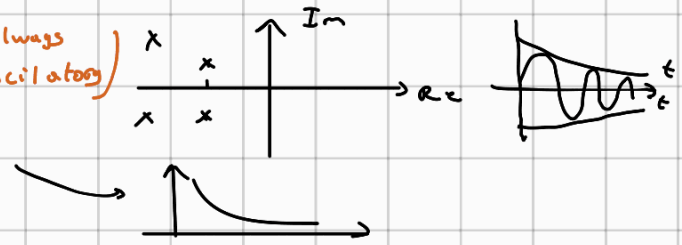
$$= \omega_n\left(-\zeta \pm \sqrt{\zeta^2 - 1}\right)$$

Underdamped Systems

if $\zeta < 1$ $\Rightarrow$ complex conjugate poles (always oscillatory)

critically damped systems

(boring) if $\zeta = 1$ $\Rightarrow$ repeated real valued poles



if $\zeta > 1$ $\Rightarrow$ seperate and real valued poles overdamped system

Homework (homogen solution)

Response due to external excitation (particular solution)

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Specify the input

I) Impulse excitation $R(s) = 1$

case I) $\zeta < 1$

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{(s + \zeta\omega_n - i\omega_d)(s + \zeta\omega_n + i\omega_d)}$$

$$= \frac{A}{s + \zeta\omega_n - i\omega_d} + \frac{B}{s + \zeta\omega_n + i\omega_d}$$

$$s_{1,2} = +\omega_n\left(-\zeta \pm \sqrt{\zeta^2 - 1}\right)$$

$$= \omega_n\left(-\zeta \pm i\sqrt{1 - \zeta^2}\right)$$

$$= -\zeta\omega_n \pm i\,\omega_n\sqrt{1 - \zeta^2}$$

damped natural freq $= \omega_d$

The response frequency for damped systems is ω_d

$$\frac{C(s)}{R(s)} = \frac{A}{s + \zeta\omega_n - i\omega_d} + \frac{B}{s + \zeta\omega_n + i\omega_d} \Rightarrow c(t)$$

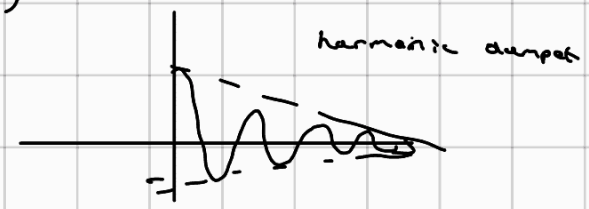
$$A = \frac{\omega_n^2}{s + \zeta\omega_n + i\omega_d}\Bigg|_{s = -\zeta\omega_n + i\omega_d}$$

$$B = \left(\frac{\omega_n^2}{s+\zeta\omega_n+i\omega_d} \right)_{s=-\zeta\omega_n-i\omega_d}$$

$$c(t) = A\, e^{(-\zeta\omega_n+i\omega_d)t} + B\, e^{(-\zeta\omega_n-i\omega_d)t}$$

$$= e^{-\zeta\omega_n t}\left(A\, e^{i\omega_d t} + B\, e^{-i\omega_d t} \right)$$

$$= \alpha\, e^{-\zeta\omega_n t} \cos\omega_d t$$



$$\begin{cases} e^{i\theta} = \cos\theta + i\sin\theta \\ e^{-i\theta} = \cos\theta - i\sin\theta \end{cases} \rightarrow \frac{e^{i\theta}+e^{-i\theta}}{2} = \cos\theta$$

case II) $\zeta = 1$; $s_{1,2} = -\zeta\omega_n$

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{(s+\zeta\omega_n)^2} = \frac{\hat{A}}{s+\zeta\omega_n} + \frac{\hat{B}}{(s+\zeta\omega_n)^2}$$

$$\hat{B} = (s+\omega_n)^2 \left(\frac{\omega_n^2}{(s+\zeta\omega_n)^2} \right)\Big|_{s=-\zeta\omega_n}$$

$$\hat{A} = \frac{d}{ds}\left(\frac{\omega_n^2}{s+\zeta\omega_n} \right)\Big|_{s=-\zeta\omega_n}$$

critical damping

$$c(t) = A\, e^{-\zeta\omega_n t} + \hat{B}\, t\, e^{-\zeta\omega_n t}$$

$$= e^{-\zeta\omega_n t}\left(\hat{A} + \hat{B} t \right)$$

III) ($\zeta > 1$)

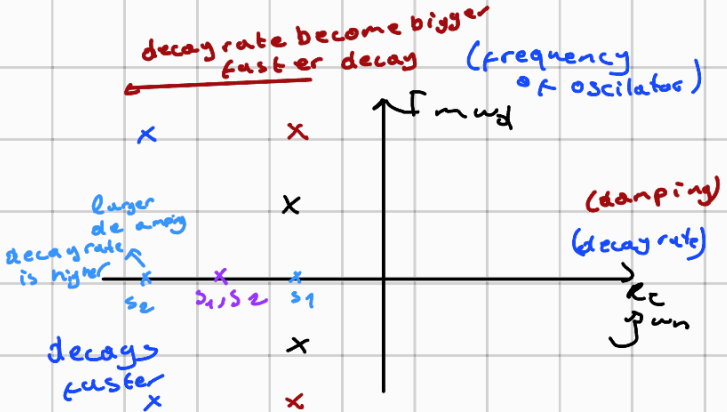
$$c(t) = \hat{A}\, e^{\left(-\zeta-\sqrt{\zeta^2-1}\right)\omega_n t} + \hat{B}\, e^{\left(-\zeta+\sqrt{\zeta^2-1}\right)\omega_n t}$$

2nd order systems

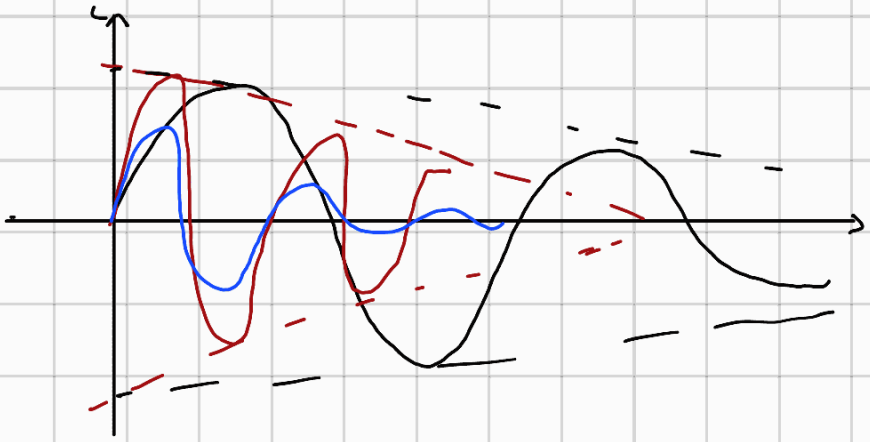
$\zeta < 1$ $\quad s_{1,2} = -\zeta\omega_n \pm i\omega_d$ ($-\zeta\omega_n$: Re, ω_d: Im)

$\zeta = 1$ $\quad s_{1,2} = -\zeta\omega_n$ (repeated) Re

$\zeta > 1$ $\quad s_{1,2} = -\zeta\omega_n \pm \sqrt{\zeta^2-1}$ (Re distinct)



If poles are C C → response is oscillatory and decaying



for ζ $\omega_d = 0$ no oscilating frequono

decodc al thesanctime

for $\zeta > 1$

$s_1 = -\zeta\omega_n + \sqrt{\zeta^2 - 1}$

$s_2 = -\zeta\omega_n - \sqrt{\zeta^2 - 1}$

$C(t) = \underbrace{A e^{s_1 t}}_{\checkmark} + B e^{s_2 t}$ → decay fastly

this is the most important part

Step response calculation

(just for $\zeta < 1$)

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$R(s) = \frac{1}{s}$$

$$C(s) = \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} = \frac{A}{s} + \frac{Bs + C}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$= \frac{1}{s} - \frac{s + 2\zeta\omega_n}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{1}{s} - \frac{s + \zeta\omega_n}{(s + \zeta\omega_n)^2 + \omega_n^2} - \frac{\zeta\omega_n}{(s + \zeta\omega_n)^2 + \omega_n^2}$$

$\frac{\zeta}{\sqrt{1-\zeta^2}}$ ω_n $\quad$ $\frac{\sqrt{1-\zeta^2}}{\sqrt{1-\zeta^2}}$

$$\mathcal{L}\{e^{-at}\sin\omega t\} = \frac{\omega}{(s+a)^2 + \omega^2}$$

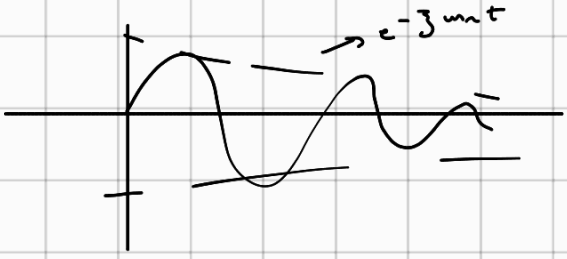
$\omega_d = \omega_n\sqrt{1-\zeta^2}$

$$\mathcal{L}[e^{-a t}\sin\omega t] = \frac{s + a}{(s+a)^2 + \omega^2}$$

$$c(t) = \mathcal{L}^{-1}[C(s)] = 1 - e^{-\zeta\omega_n t}\cos\omega_d t - \frac{\zeta e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin\omega_d t$$

$$= 1 - e^{-\zeta\omega_n t}\left(\cos\omega_d t + \frac{\zeta}{\sqrt{1-\zeta}}\sin\omega_d t\right)$$

$$c(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin\left(\omega_d t + \tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}\right)$$



$$e(t) = r(t) - c(t) = \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\sin(\omega_d t + \phi)$$